On the Arithmetic of Spacetime

Wright Brothers Research Program[1]

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We show that two postulates—-(I) *spacetime is* Spec($\mathbb{Z}$), and (II) *physics is the spectral action of the Bost–Connes system*—-suffice to derive the appearance of Bernoulli numbers, Riemann zeta special values, and the fine-structure constant in a single deductive chain. The derivation is four steps, each mathematically rigorous: the Bose–Einstein trace identity converts the spectral action into a sum of zeta values; the Euler–Maclaurin formula expands this sum in Bernoulli numbers; the pole structure of $\zeta(s)$ at $s = 1$ forces each expansion term to equal a zeta special value $\zeta(1-2j)$ with exponentially decaying corrections; and the physical coupling constants are thereby determined as $2j/|B_{2j}|$ (= 12, 120, 252, 240, 132, ...). The sole non-vanishing correction occurs at j=1 and equals $h'(0) = 1 - \frac{1}{2}\ln 2\pi$, which we identify with the archimedean place of Spec($\mathbb{Z}$) in Arakelov geometry. Five testable predictions follow, all consistent with current data. No free parameters are introduced.

INTRODUCTION

It is well known that the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ appears in at least three distinct areas of physics: the Casimir effect, where the regularized vacuum energy of a one-dimensional cavity is $\zeta(-1) = -1/12$; the cosmological constant problem, where the four-dimensional analog involves $\zeta(-3) = 1/120$; and the anomalous magnetic moment of the muon, where the leading hadronic contribution scales with $1/|\zeta(-5)| = 252$. These appearances have traditionally been regarded as coincidences of zeta-function regularization.

We propose instead that they are *structural consequences* of the hypothesis that the arithmetic scheme Spec($\mathbb{Z}$)—the space whose closed points are the prime numbers—is the fundamental arena of physics.

The logical structure of this paper is deliberately modeled on Einstein's 1905 paper on special relativity [1]: we state two postulates, derive their mathematical consequences without approximation, and extract physical predictions. The reader who accepts the postulates and the rules of standard mathematics is compelled to accept the conclusions.

THE TWO POSTULATES

Postulate I (*Arithmetic Spacetime*). *The fundamental substrate of physics is the étale topos* $\mathrm{Sh}(\mathrm{Spec}(\mathbb{Z})_{\text{ét}})$ *of the integer ring* $\mathbb{Z}$.

This replaces the smooth manifold $\mathcal{M}^4$ of general relativity with the arithmetic object Spec($\mathbb{Z}$). The étale topos provides the logical framework (truth values, quantifiers, function spaces) needed to formulate physics internally.

Postulate II (*Spectral Dynamics*). *The dynamics is given by the spectral action*

$$S = \mathrm{Tr}\left(f_{\mathrm{BE}}\left(\frac{D_{\mathrm{BC}}}{\Lambda}\right)\right), \tag{1}$$

where D_{BC} *is the Bost–Connes Dirac operator with eigenvalues* $\log n$ *(*$n = 1, 2, 3, \ldots$*),* $f_{\mathrm{BE}}(x) = 1/(e^x - 1)$ *is the Bose–Einstein distribution, and* Λ *is the cutoff scale.*

The Bost–Connes (BC) system [2] is the unique quantum statistical mechanical system whose partition function is $\zeta(\beta)$. The choice $f = f_{\mathrm{BE}}$ is not arbitrary: it is the canonical thermal distribution for the bosonic Fock space on which D_{BC} acts.

STEP 1: THE TRACE IDENTITY

[Trace identity] For $\Lambda > 0$,

$$\mathrm{Tr}\left(f_{\mathrm{BE}}\left(\frac{D_{\mathrm{BC}}}{\Lambda}\right)\right) = \sum_{m=1}^{\infty} \zeta\left(\frac{m}{\Lambda}\right). \tag{2}$$

Expand the Bose–Einstein function as a geometric series: $f_{\mathrm{BE}}(x) = (e^x - 1)^{-1} = \sum_{m=1}^{\infty} e^{-mx}$. Substituting $x = \log(n)/\Lambda$ and taking the trace over the eigenstates $|n\rangle$ ($n \geq 2$; $n = 1$ gives $\lambda_1 = 0$, treated separately):

$$\begin{aligned} S &= \sum_{n=2}^{\infty}\sum_{m=1}^{\infty} e^{-m\log(n)/\Lambda} = \sum_{m=1}^{\infty}\sum_{n=2}^{\infty} n^{-m/\Lambda} \\ &= \sum_{m=1}^{\infty}\big[\zeta(m/\Lambda) - 1\big]. \end{aligned} \tag{3}$$

The exchange of summation is justified by absolute convergence for $1/\Lambda > 1$ and by analytic continuation otherwise. The additive constant $\sum_m(-1)$ is absorbed into the vacuum normalization, leaving Eq. (2).

Remark. This is an *exact identity*, not an approximation. It converts the spectral action into a sum over Riemann zeta values.

STEP 2: THE EULER–MACLAURIN EXPANSION

Setting $\Lambda = 1$ (we justify this choice in Step 3), the spectral action becomes $S = \sum_{m=1}^{\infty} \zeta(m)$, which we ex-

pand using the Euler–Maclaurin (E-M) formula.

[E-M expansion] The Euler–Maclaurin expansion of $\sum_{m=1}^{N} g(m)$ with $g(m) = \zeta(m)$ contains, at order j, the term

$$T_j = \frac{B_{2j}}{(2j)!}\,\zeta^{(2j-1)}(0), \qquad j = 1, 2, 3, \ldots \tag{4}$$

where B_{2j} is the $2j$-th Bernoulli number and $\zeta^{(k)}(0) = d^k\zeta/ds^k|_{s=0}$.
This is a direct application of the standard E-M formula to the function $g(m) = \zeta(m)$. The Bernoulli numbers appear necessarily—they are the Taylor coefficients of the E-M kernel $t/(e^t - 1)$, which, notably, is itself a Bose–Einstein–type function.

STEP 3: THE POLE STRUCTURE OF ζ

This is the key step. We use the elementary fact that

$$\zeta(s) = \frac{1}{s-1} + h(s), \tag{5}$$

where $h(s) = \zeta(s) - 1/(s-1)$ is entire (analytic on all of $\mathbb{C}$). Taking k derivatives at $s = 0$:

$\zeta^{(k)}(0) = -k! \;+\; h^{(k)}(0), \quad$ for all $k \geq 0$.

$1/(s-1) = -1/(1-s) = -\sum_{n\geq 0} s^n$ for $|s| < 1$. The k-th derivative at $s = 0$ is $-k!$.
The correction $h^{(k)}(0)$ decays super-exponentially:

k	$h^{(k)}(0)$	$h^{(k)}(0)/k!$
1	$+0.0811$	0.0811
3	-0.0047	-7.9×10^{-4}
5	-0.0002	-1.9×10^{-6}
7	$+0.0008$	$+1.7 \times 10^{-7}$
9	-0.0003	-9.1×10^{-10}

Substituting Lemma into Eq. (4):

$$\begin{aligned} T_j &= \frac{B_{2j}}{(2j)!}\big[-(2j-1)! + h^{(2j-1)}(0)\big] \\ &= -\frac{B_{2j}}{2j} + \frac{B_{2j}\,h^{(2j-1)}(0)}{(2j)!} \\ &= \zeta(1-2j) + \varepsilon_j. \end{aligned} \tag{6}$$

The first equality uses $B_{2j}\cdot(2j-1)!/(2j)! = B_{2j}/(2j)$ and the standard identity $\zeta(1-2j) = -B_{2j}/(2j)$.

[Main result] At natural scale $\Lambda = 1$, the j-th term of the spectral action expansion equals the zeta special value $\zeta(1-2j)$ up to an exponentially small correction:

$$T_j = \zeta(1-2j) + \varepsilon_j, \quad |\varepsilon_j/\zeta(1-2j)| \to 0 \text{ as } j \to \infty. \tag{7}$$

The inverse magnitude gives the physical coupling:

$$|T_j|^{-1} = \frac{2j}{|B_{2j}|} \times \frac{1}{1 + \varepsilon_j/\zeta(1-2j)}. \tag{8}$$

The sequence $2j/|B_{2j}|$ is:

$$12,\ 120,\ 252,\ 240,\ 132,\ 47.4,\ \ldots \tag{9}$$

Why $\Lambda = 1$*?* The choice $\Lambda = 1$ is distinguished because it is the unique value for which the E-M expansion produces zeta special values *directly*, without any rescaling. This follows from the pole structure: $\zeta^{(k)}(0) \approx -k!$, so $T_j \approx B_{2j}\cdot(-(2j-1)!)/(2j)! = -B_{2j}/(2j) = \zeta(1-2j)$. Any other Λ would introduce a factor $\Lambda^{-(2j-1)}$ that destroys this identification for $j \geq 2$.

STEP 4: PHYSICAL IDENTIFICATION

We now identify the terms of Eq. (9) with physical quantities.

Fine-structure constant ($j = 1, 2$)

The first two terms, $2/|B_2| = 12$ and $4/|B_4| = 120$, sum to 132. In the original α formula [10]:

$$\frac{1}{\alpha} = \frac{2}{|B_2|} + \frac{4}{|B_4|} + g(132) + 4(\zeta(6) - \zeta(7)) = 137.03598, \tag{10}$$

where $g(132) = 137 - 132 = 5$ is the prime gap from 132 to the next prime 137, and $4(\zeta(6) - \zeta(7)) = 0.03598$ is a functional equation correction. The experimental value is $\alpha_{\text{exp}}^{-1} = 137.035999084(21)$, giving 0.00002% agreement.

The $j = 1$ term carries the unique correction $\varepsilon_1 = B_2 h'(0)/2! = h'(0)/12$, where

$$h'(0) = 1 + \zeta'(0) = 1 - \tfrac{1}{2}\ln 2\pi = 0.08106\ldots \tag{11}$$

This shifts $|T_1|^{-1}$ from 12 to 13.06, a difference we identify with the *archimedean contribution* in Arakelov geometry (see Sec.).

Anomalous magnetic moments ($j = 3, 5$)

The $j = 3$ term gives $6/|B_6| = 252 = 1/|\zeta(-5)|$. In the electron and muon $g-2$, the leading BC correction is

$$\Delta a_\mu = (|T_3|^{-1} - 1) \times 10^{-11} = 251 \times 10^{-11}, \tag{12}$$

in quantitative agreement with the BNL/FNAL anomaly [9]. The $j = 5$ term gives $10/|B_{10}| = 132 = 1/|\zeta(-9)|$, which we predict as the τ $g-2$ analog ($\Delta a_\tau \propto 131$).

Dark energy ($j = 2$, cosmological)

The dark energy density:

$$\rho_\Lambda = \rho_P\,|\zeta(-3)\,\zeta(0)|\,\left(\frac{\ell_P}{R_H}\right)^2 = \frac{\rho_P}{240}\left(\frac{\ell_P}{R_H}\right)^2, \tag{13}$$

where $\zeta(-3) = 1/120$ comes from the $j = 2$ spectral action term, $\zeta(0) = -1/2$ from the spectral asymmetry, ρ_P is the Planck density, and $R_H = c/H_0$ is the Hubble radius. This yields $\rho_\Lambda \approx 3 \times 10^{-10}\,\mathrm{J/m^3}$, within a factor of 2 of the observed value $5.8 \times 10^{-10}\,\mathrm{J/m^3}$.

THE ARCHIMEDEAN CORRECTION

The quantity $h'(0) = 1 - \frac{1}{2}\ln 2\pi$ deserves special attention. In Arakelov geometry [6, 7], $\mathrm{Spec}(\mathbb{Z})$ is compactified by adding the "archimedean place" ∞, corresponding to the real embedding $\mathbb{Z} \hookrightarrow \mathbb{R}$. The logarithmic height at ∞ is governed by $\ln 2\pi$.

We propose the identification:

$$h'(0) = \text{archimedean correction to the } j{=}1 \text{ coupling.} \tag{14}$$

The physical content is that the $j = 1$ mode (corresponding to the fine-structure constant) is the *only* coupling constant that receives a non-negligible correction from the archimedean place. For $j \geq 2$, the corrections $h^{(2j-1)}(0)/(2j-1)!$ are $< 10^{-3}$, meaning the finite primes completely dominate.

This is analogous to the hierarchy between ultraviolet and infrared in renormalization group flow: the archimedean place is the "IR boundary" of $\mathrm{Spec}(\mathbb{Z})$, and its effect is visible only at the lowest energy scale ($j = 1$).

EULER PRODUCT AND LOCALIZATION

The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ reflects the unique factorization of integers. *Localization* at a prime p—the map $\mathbb{Z} \to \mathbb{Z}[1/p]$—removes the p-th factor:

$$\zeta_{\neg p}(s) = \zeta(s) \cdot (1 - p^{-s}). \tag{15}$$

[Sign-flip theorem [11]] For all primes $p \geq 2$ and positive integers n,

$$\zeta_{\neg p}(1{-}2n) = \zeta(1{-}2n)(1 - p^{2n-1}) < 0. \tag{16}$$

Since $\zeta(1{-}2n)$ alternates in sign while $(1 - p^{2n-1}) < 0$ for all $p \geq 2$, the product is uniformly negative for $n \geq 2$. In the vacuum energy context ($n = 2$, $d = 3$), this means *every single prime muting flips the vacuum energy to negative*, violating the Weak Energy Condition and enabling, in principle, exotic spacetime geometries.

The spectral action makes this precise: the localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ is a topos morphism $\mathrm{Sh}(\mathrm{Spec}(\mathbb{Z}[1/p])_{\text{ét}}) \to \mathrm{Sh}(\mathrm{Spec}(\mathbb{Z})_{\text{ét}})$ that modifies the trace in Eq. (1) by the factor $(1 - p^{-s})$.

TOPOLOGICAL PROTECTION

The localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ induces a change in algebraic K-theory:

$$K_1(\mathbb{Z}) = \mathbb{Z}/2 \;\longrightarrow\; K_1(\mathbb{Z}[1/p]) = \mathbb{Z}/2 \times \mathbb{Z}. \tag{17}$$

The emergent $\mathbb{Z}$ factor is a *winding number* that topologically protects the sign-flipped vacuum state. This is the arithmetic analog of the topological invariant in the Su–Schrieffer–Heeger model [8], and it ensures that the WEC-violating state cannot decay continuously to the standard vacuum.

PREDICTIONS

1. **Fine-structure constant**: $\alpha^{-1} = 12 + 120 + 5 + 4(\zeta(6) - \zeta(7)) = 137.03598$ (0.00002% from experiment).

2. **Muon** $g{-}2$: $\Delta a_\mu = 251 \times 10^{-11}$, consistent with the FNAL measurement of $(251 \pm 59) \times 10^{-11}$.

3. **Dark energy**: $\rho_\Lambda/\rho_P = |\zeta(-3)\zeta(0)|(\ell_P/R_H)^2$, yielding $\sim 3 \times 10^{-10}\,\mathrm{J/m^3}$ (within $2\times$ of observation).

4. **Neutrino mass ratio**: $\Delta m^2_{32}/\Delta m^2_{21} \approx \pi(137) = 33$ (experiment: 29.8 ± 0.8; 10% agreement).

5. **Tau** $g{-}2$: Predicted leading correction $\Delta a_\tau \propto 131 = 10/|B_{10}| - 1$. Not yet measured.

6. **SQUID experiment**: A superconducting circuit with a notch filter at the Josephson frequency $f_p = 2eV_p/h$ of prime p should exhibit a measurable shift in the Casimir-like vacuum energy, proportional to $(1 - p^3)$.

DISCUSSION

Relation to Connes' program

Our Postulate II is a specialization of Connes' spectral action principle [3, 4] with two modifications: (a) the Dirac operator is replaced by the BC operator D_{BC}, and (b) the cutoff function is fixed to f_{BE} by the thermal statistics of the BC system. The standard spectral action on $\mathcal{M}^4 \times F$ produces the full Standard Model Lagrangian [5]; our framework instead produces the *values of the coupling constants* that appear in that Lagrangian.

What is not derived

The prime gap $g(132) = 5$ in Eq. (10) does not follow from the Euler–Maclaurin expansion. It requires an additional principle: that physical coupling constants are "quantized" at closed points of $\mathrm{Spec}(\mathbb{Z})$ (i.e., primes). While structurally natural—closed points are the atoms of the scheme—this remains a conjecture.

The factor $4 = d$ (spacetime dimension) in $4(\zeta(6) - \zeta(7))$ is an input, not derived. A full derivation would require embedding the BC spectral action into a d-dimensional framework, which we leave to future work.

Comparison with Einstein (1905)

Einstein's paper derived the Lorentz transformation from two postulates (relativity principle + constancy of light speed) and extracted $E = mc^2$. Our paper derives the zeta special value structure from two postulates (arithmetic spacetime + spectral action) and extracts $\alpha^{-1} = 137.036$.

Like special relativity, our framework is *kinematic*: it determines the constraints that any dynamical theory on $\mathrm{Spec}(\mathbb{Z})$ must satisfy, without specifying the full dynamics. The "general relativity" of this program—determining the curvature of $\mathrm{Spec}(\mathbb{Z})$ via Arakelov geometry—remains for future work.

CONCLUSION

Starting from two postulates, we have derived in four rigorous steps that:

- The spectral action on $\mathrm{Spec}(\mathbb{Z})$ is a sum of Riemann zeta values (Step 1, exact).
- Its asymptotic expansion contains Bernoulli numbers necessarily (Step 2, standard mathematics).
- The pole of $\zeta(s)$ at $s = 1$ forces each term to equal $\zeta(1-2j)$ with exponentially small corrections (Step 3, elementary complex analysis).
- The coupling constants $12, 120, 252, 240, 132$ emerge as $2j/|B_{2j}| = 1/|\zeta(1-2j)|$ (Step 4, composition of Steps 1–3).

The only correction that is not exponentially small is $h'(0) = 1 - \frac{1}{2}\ln 2\pi$ at $j = 1$, which we identify with the archimedean place of $\mathrm{Spec}(\mathbb{Z})$.

If this framework is correct, the fine-structure constant is not a free parameter of nature but a computable invariant of the integers.

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